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Abstract
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1. Introduction

1.1. Context and Motivation (for VR exergaming)

Exergaming, short for exertion gaming, is defined as technology-driven physical activities that require participants to be exerting themselves in one way or another in order to play a game. Exergaming was initially conceptualised in the late 1980s and gained significant popularity with the introduction of games like “Dance Dance Revolution” in the late 1990s. The introduction of the Nintendo Wii and Xbox Kinect in the early 2000s further pushed exergaming into the mainstream media, with these consoles and their games demonstrating the potential of exercise in gaming. The term “exergaming” entered the Collins English Dictionary in 2007, marking a point where the concept was officially recognised and popularised. Exergaming has continued to develop into not only video games, but also gaming technologies integrated with exercise equipment, mobile apps, and virtual reality.

Virtual reality has become a compelling platform for exergaming development due to its unparalleled immersion and captivating nature. VR in exergames has shown to increase both enjoyment and motivation of physical exercise over both short [Farrow et al, 2019], and long periods of time [Michael & Lutteroth, 2020]. Virtual reality allows for a wider range of virtual activities compared to flatscreen exergaming, including sports, dance, and adventure games. Additionally, VR environments are promising treatment to mitigate functional decline and promote cognitive rehabilitation [Marotta et al, 2022]. Some studies suggest that VR exergames can have a positive impact on cognitive function, memory, and depression, particularly in older adults {FIND REFERENCE}.

1.2. Problem Statement (lack of adaptive, engaging VR exergames leveraging social dynamics (competition/cooperation).

Despite the growing popularity of VR exergaming as a tool to promote physical exercise, many VR fitness applications lack dynamic social interaction implementations that effectively sustain user motivation and engagement, while research has shown that incorporating competition and cooperation in exergames can both enhance user performance and effort [FIND REFERENCE], we have a limited understanding of how these social dynamics specifically impact user experience within VR cycling. Furthermore, more existing VR exergames are not tailored towards the user’s specific fitness level, resulting in an experience that may be either too difficult or too easy, reducing long-term engagement.

Therefore, there is a critical need for further exploration on how competitive and cooperative elements in VR cycling applications influence the user’s performance, motivation, and engagement across a diverse range of users. Additionally, the integration of adaptive AI opponents and teammates that respond to the user’s effort could provide a scalable solution for tailoring difficulty and social interaction, making VR exergaming more accessible and enjoyable. Addressing these gaps is essential for developing VR fitness solutions that are immersive, personalised, and sustainable in promoting physical exercise.

1.3. Aims of the project

The aim of this project is to develop a VR cycling game with varied social interaction designs to investigate how cooperative and competitive gameplay mechanics influence user performance, engagement, and fitness outcomes. The project seeks to address the lack of accessible and engaging VR cycling platforms that cater to users with different motivational triggers and fitness levels. By integrating adaptive AI-driven scenarios and real-time feedback, the project aims to create an exergaming experience that is both engaging and inclusive for a wide range of users.

This project aims to develop three distinct cycling scenarios. These scenarios will take place in a prototype velodrome with minimal visual clutter to distract from the workout with the user completing the same physical workout routine through each scenario. (1) A baseline scenario involving the user pedalling around the velodrome on a solo bike ride, (2) a cooperative scenario involving the user pedalling around in a paceline with three AI teammates, (3) a competitive scenario involving the user pedalling around in a race competing against AI's with performance based on the users previous two scenario performances, with relative difficulty variating based on the users current heart rate.

This project aims to incorporate social interaction and gamification elements into VR exergaming through implementing features such as: real-time performance tracking for the user to view if desired as well as leaderboard performances based on previous bests, AI teammates and opponents capable of dynamic behaviour adjustments based on user effort and exertion to maintain challenge and engagement, post-race analytics to provide users with detailed performance feedback across all scenarios.

This project aims to evaluate user performance and engagement through a mixed-method user study to assess performance metrics such as speed, cadence, power output, heart rate, and time-to-completion across the different scenarios. Additionally, this project aims to gather qualitative feedback through surveys and questionnaires to evaluate user engagement, motivation, and experience in each scenario, as well as investigate possible correlations between user personality traits and preferences for competitive or cooperative gameplay modes.

This project aims to ensure accessibility and usability of the system, with user interfaces designed in an intuitive manner to optimise ease of use. The implementation of adaptive AI difficulty scaling, to tailor challenge levels to users of varying levels of fitness, as well as follow UI/UX best practices to ensure users can navigate the system effectively without unnecessary complexity or barriers.

1.4. Intended Beneficiaries

General fitness enthusiasts and individuals seeking motivational exercise tools. For people looking for more engaging ways to exercise, especially those who struggle with traditional fitness routines due to lack of motivation or boredom, VR exergaming can be a great solution through its gamification of real-world exercises. Individuals of varying fitness levels, from beginners to advanced athletes, who can benefit from adaptive AI that tailor's difficulty to their abilities.

Individuals interested in home-based or VR-based fitness solutions. Home exercisers, especially those with limited access to gyms or outdoor activities, who are seeking immersive and interactive fitness experiences. People motivated by interactive environments and social gaming elements to maintain regular exercise habits.

Health and rehabilitation professionals. Physical therapists and rehabilitation specialists looking for innovative ways to encourage cardiovascular exercise and rehabilitation training through VR. Could be adapted for patients needing low impact yet engaging cardio sessions in controlled environments.

Researchers and developers in VR, exergaming, and health technology. Academic researchers exploring the psychological and physiological impacts of social dynamics in exergames, such as cooperation and competition. Game developers and VR designers focusing on serious games for health, who can use insights from this project to create more adaptive and motivational exergaming experiences.

Organisations focused on public health and fitness promotion. Public health bodies, fitness organisations, and wellness programs aiming to promote active lifestyles using innovative technologies like VR. These organisations could leverage motivational insights from this project to design campaigns or interventions encouraging physical activity through gamified VR experiences.

The VR and gaming industry. VR developers seeking to expand into the fitness and wellness market with products that combine entertainment with physical benefits. Companies working on hardware that could integrate social and adaptive features into their platforms.

1.5. Project Scope and Limitations

This paper focuses on the development and evaluation of a VR cycling exergame designed to explore the impact of social dynamics on user engagement, motivation, and performance. The study investigates how adaptive AI teammates and opponents can enhance the VR exergaming experience and maintain player motivation through real-time difficulty adjustments based on effort and exertion levels.

The scope of this project includes:

- VR Cycling Simulation – Development of a virtual velodrome cycling environment with three distinct gameplay modes: Baseline, Cooperative, Competitive. With clear UX design allowing users to navigate to their desired experience.
- Social Interaction Elements – Implementation of AI-driven teammates and opponents that adapt dynamically to the user's performance.
- Engagement & Performance Assessment – Evaluation of user motivation, effort, and fitness metrics (speed, cadence, heartrate, power output) through quantitative and qualitative methods.
- Gamification & UI/UX – Design of intuitive interfaces, real-time performance tracking, and post session analytics to enhance user experience.
- Adaptive Difficulty Scaling- Integration of AI-based adjustments to tailor challenge levels according to user fitness levels and exertion.

Project Limitations:

- Limited Participant Pool – The study involved in this project does not account for all demographics, such as holder adults or individuals with disabilities.
- Short-term Study Duration – The research evaluates immediate engagement and performance rather than long-term fitness adherence.
- Single-Modality Exergame – The focus is solely on VR cycling, without exploring other exercise forms such as running or strength training.
- Absence of Multiplayer Mode – While the study explores AI-driven social interaction, real-time multiplayer experiences may be more effective, however are beyond the scope of this project.

- Absence of Environmental Modifications – This project revolves solely around a velodrome map prototype, and does not take into account how changes in environment and surroundings impacts engagement, motivation, and performance.

1.6. Summary of Important Outcomes

This project provides valuable insights into the role of social dynamics in VR exergaming, specifically in the context of Cycling. The key outcomes of this paper include:

2. Background

2.1. VR exergaming landscape (review key studies e.g. shaw et al, Michael et al)

Competition and Cooperation in Exergaming

Exergaming is a field which has gained traction in research over the last 10-15 years. Studies have suggested that exergames have the potential to motivate inactive individuals to exercise (Warburton et al, 2007), but that they can also struggle with prolonged adherence similar to conventional exercise (Mestre, Dagonneau & Mercier, 2011). The focus on this project, however, is to see how changes in social dynamics in cooperative and competitive environments impact exercise adherence, enjoyment, motivation, and performance, and to additionally see if results correlate with particular personality types favouring different dynamics.

Previous studies show that competitive and cooperative factors in exergames influence user motivation. Peng & Crouse (2013) used an exergame possessing three different scenarios, (1) A single player mode where users played against their self pretest score; (2) a cooperative mode with another player in the same physical space; (3) a mode with another player in parallel competition in separate physical spaces. In this study they found (3) parallel competition to be the optimal mode, resulting in high enjoyment and future play motivation and high physical intensity. The study also found that cooperation was more enjoyable and motivating than solitary play, but the singleplayer mode led to greater levels of physical exertion.

Competition can have a strong impact on intrinsic motivation (Song et al 2009), a study that explored both highly and lowly competitive individuals in both highly and lowly competitive conditions found that intrinsic motivation could be increased for highly competitive individuals when presented with competitive exergaming settings, however that for less competitive individuals being placed in a competitive environment suffered detrimental effects to their exercise experience, reducing their enjoyment and their likelihood of engaging in further sessions.

Cooperative dynamics have also exhibited benefits in the field of exergaming. A study on Adolescent exergame play for weight loss and psychosocial improvement (Staiano, Abraham & Calvert (2013) had participants from a public high school randomly assigned to different modes (competitive, cooperative, control) within the Nintendo Wii Active game for 30-60 minutes. Results showed that players in the cooperative exergame lost significantly more weight than the control group which didn't lose any weight. The competitive players did not differ significantly from the other conditions. It was also found that cooperative players significantly increased in self-efficacy compared to the control group and both competitive and cooperative players significantly increased in peer support. This concluding that exergames, particularly in cooperative settings can be an effective tool for weight loss among youth.

Recent cross-sectional studies further examined the effects of competition and cooperation on exercise performance in immersive virtual reality cycle exergames (shaw et al 2016). In one study, competing against a ghost of the user's previous session resulted in a greater distance travelled by the user compared to cycling with a virtual trainer present. Additionally, the ghost mode was rated as more enjoyable than both solo cycling and cycling with a virtual trainer. A follow-up study revealed that a competitive virtual trainer system elicited greater distance travelled and caloric expenditure compared to a cooperative virtual trainer system and was rated as more motivating. Interestingly, in both studies, self-reported enjoyment and motivation did not significantly correlate with actual performance metrics (distance travelled, and calories burned). These findings reinforce the effectiveness of competitive elements in exergaming as tools to elicit higher levels of exercise, demonstrating that competitors can successfully substitute for human opponents in driving improved performance.

Interactive Feedforward in Exergaming

Interactive feedforward is an interactive adaptation of the psychological feedforward training method where rapid improvements in performance are achieved by creating self-models showing previously unachieved performance levels (Barathi et al 2020). It can be used to enhance performance, understanding, and engagement by providing users with previous of potential outcomes and guiding them towards better results.

Unlike traditional feedback, which focuses on past performance, feedforward focuses on future possibilities and potential outcomes. Its about anticipating what might happen and guiding users toward desired results. Interactive feedforward takes this concept further by incorporating real-time interaction and feedback within the system. This allows for dynamic adjustments and improvements as the user progresses.

A HIIT (High Intensity Interval Training) 4-week longitudinal study on exergaming with the use of interactive feedforward had user's complete workouts with no ghost condition, and with a ghost condition where logged performances were put into a ghost non-player character with an increased performance of 5% to visualise the future potential performance of the user. During this study participants who engaged in the non-competitive scenario made an average performance gain of 8%, compared to the participants who engaged in the competitive scenario using interactive feedforward made an average performance gain of 16%. Additionally, participants in the competitive (Multi-Ghost) group found that they were more intrinsically motivated, had a greater perceived competence, and but in a greater amount of effort, compared to the solo cycling group (Non-Competitive) (Michael et al 2020). Overall, this study proved that interactive feedforward can be an effective method of improving player performance over time.

A study in 2018 examined the application of interactive feedforward in high-intensity VR exergaming, addressing the gap in effective protocols for high intensity exercise in virtual environments. Researchers tested the interactive feedforward training technique by having participants cycle against either models of their own selves' previous performances, or virtual competitors, with varying levels of resistance and awareness. The results demonstrated that interactive feedforward successfully enhanced participant performance while preserving intrinsic motivation, even when participants were aware of the interventions. Notably, competing against modified self-models proved more effective than competing against virtual competitors. These findings suggest that interactive feedforward offers a promising approach for designing high-intensity VR exergames that can simultaneously improve physical performance and maintain user engagement (Barathi et al 2018).

2.2. Effects of a lack of exercise

Physical Health Consequences

Physical inactivity is a primary cause of most chronic diseases (Booth et al 2012). In 2022 31% of adults worldwide (1.8 billion people), did not meet the recommended levels of physical activity []. This is an increase of 5 percent points between 2010 and 2022. If this trend continues, the proportion of adults not meeting recommended levels of physical activity is projected to rise to 35% by 2030 []. Physical inactivity puts adults at greater risk of cardiovascular diseases such as heart attacks and strokes, type 2 diabetes, dementia and cancers such as breast and colon. Additionally, physical inactivity can cause physiologic muscle atrophy as people's muscles aren't being used enough. This is particularly a problem for those who have seated jobs [], and considering it is claimed 81% of UK office workers spend between four and nine hours each day sitting at their desk [], this is a considerable concern for a large amount of the population.

Mental Health Impacts

Mental illness has become a major health problem worldwide, particularly during COVID-19 it was estimated that approximately 12% of the global population had various mental illnesses, which was associated with a significant burden on healthcare systems like the NHS translating into approximately 5% of disability-adjusted life years lost (Schuch et al 2021). This research found compelling evidence that physical activity and exercise can prevent common mental disorders, such as depression and anxiety disorders, and have multiple beneficial factors on the physical and mental health of people who already have mental health disorders.

Societal and Economic Costs

According to the World Health Organization (WHO), 63% of worldwide deaths in 2008 were caused by major preventable diseases []. In 2011, the cost of chronic disease worldwide was estimated to reach $47 trillion by 2030 []. Physical inactivity at work leads to decreased productivity and increased absenteeism due to various health issues and cognitive impairments. Reduced physical activity can cause fatigue, musculoskeletal problems, and mental health issues, all of which contribute to time off work. Additionally, a lack of physical activity can impair cognitive function, making it harder for employees to focus and concentrate, impacting their productivity while at work. In terms of presenteeism, individuals who are not physically active lose a larger amount of working time (between 2,6 and 3,71 days per year) compared to an individual who is physically active [].

The Sedentary Lifestyle Trend

The World Health Organisation (WHO) identified physical inactivity as the fourth leading risk factor for global mortality. It contributes to 6% of global deaths. Following after high blood pressure (13%), tobacco use (9%), and high blood glucose (6%) []. VR exergames provide a unique opportunity to revolutionise traditional exercise routines in ways reality simply cannot to aim to tackle the global issue of sedentary lifestyles. By merging immersive and motivating gaming experiences with pre-existing workouts, we can both increase enjoyment and motivation of exercise, as well as the convenience and accessibility of indulging in it.

2.3. Gamification and social interaction mechanics in VR fitness

In the Cambridge dictionary gamification is defined as "the practice of making activities more like games in order to make them more interesting or enjoyable" []. Gamification elements can include features such as points, challenges, levels, or badges in the contexts of exercise to increase motivation and enjoyment. Commercial VR exergames like Zwift, FitXR and Supernatural) use gamification heavily, but gamification isn't the only path to user engagement, this project focuses adopts a minimalist approach, focusing instead on performance feedback and social dynamics to increase engagement. This allows users to concentrate on exertion without visual clutter distracting them from the workout.

Rather than focusing on plain gamification, this system includes a heads-up display (HUD) for the user showing key workout metrics (speed, cadence, power, heartrate, time), which supports user engagement through self-awareness. This minimalist design aligns with the needs of users who may prefer data driven motivation rather than playful interaction.

(Shaw et al 2016) found that participants in VR cycling scenarios with cooperative and competitive social dynamics expended significantly more effort and reported higher engagement compared to those cycling alone or with a ghost. This supports the idea that social presence and dynamics can act as standalone motivators in exergames, aligning with this project's focus on AI-driven social mechanics over traditional gamified elements.

Similarly, (Michael et al 2020) demonstrated that a multighost competition model in VR cycling led to a 16% increase in performance over four weeks. This reinforces the idea that gamification is not required to drive improvement, rather, engaging challenges that evolve with the user, such as AI-based competition, can yield meaningful fitness outcomes.

According to Deci and Ryans Self-Determination Theory, intrinsic motivation arises when individuals experience competence, autonomy, and relatedness. The heads-up display (HUD) used in this system directly supports competence through real-time feedback, while the cooperative and competitive modes encourage relatedness, creating a socially meaningful workout experience even with the absence of traditional gamified features.

While Cugelman (2013) outlines game mechanics such as rewards and progress tracking as key tools in behaviour change, the paper also emphasises the importance of timely feedback and goal-related metrics, both of which are central to the program developed in this project. By presenting performance stats through a heads-up display, the user remains engaged through self-monitoring, fulfilling many of the same behavioural incentives as gamification.

2.4. Existing gaps in VR exergaming systems and preexisting solutions (focus on adaptability, dynamic challenge, user diversity).

Most VR fitness apps offer a static challenge and rarely offer intelligent adaptability for different fitness levels. This project addresses this gap by using interactive feedforward to proactively adjust AI behaviour based on historical user data which tailors the workout challenge to push the user further than they did before.

Previous studies (Shaw et al 2016) had users speed directly correlate to how fast the users cadence was, i.e. the faster you pedal the faster you go. This opens up research to flaws such as inconsistent pacing strategies across sessions where less physically active participants better learn how to prolong their workout simply by having more practice on the system. This project resolves this problem with the use of ERG mode, a feature on many smart bike trainers that automatically adjusts the resistance to help you maintain a specific power output rather than having to manually control resistance. Using ERG mode, this project was able to eliminate the variable of pacing through ensuring that the users were outputting the wattage the exercise bike was asking of them. Additionally, having participants produce a set wattage each second of the workout allows AI to better tailor the challenge to the participant as the data is smoother and more reliable when it comes to mimicking previous performances.

Past research has shown competition, and cooperation can boost motivation (shaw et al 2016, Michael et al 2020), but few systems intentionally compare these dynamics in a controlled way. Additionally, there is little analysis of user preference or personality types influencing effectiveness. This solution directly compares scenarios (1) baseline, (2) cooperative, (3) competitive, and collects both quantitative performance feedback, and qualitative form feedback after all three. From this, this project aims to test which type of social interaction is most effective and investigate if different personalities thrive in different modes.

Many fitness games assume a general audience and don't scale well from complete beginners to high performance athletes. Additionally, particularly in the commercial industry, exergames often lack inclusive design for different motivational triggers or ability levels. This project combats that using the baseline condition to calibrate difficulty, as well as emphasises accessibility in UI and game pacing.

2.5. Justification for the study based on the literature and gaps identified

The literature highlights several opportunities for further innovation within the VR exergaming space. While prior research has shown that social dynamics such as competition and cooperation enhance motivation, few systems have directly compared these interaction styles in a controlled environment, nor accounted for how personality traits influence their effectiveness. Furthermore, many existing exergames fail to dynamically adapt to varying fitness levels, resulting in static challenges that may demotivate users over time or fail to meet their needs.

This project addresses these gaps by integrating adaptive AI via interactive feedforward, implementing ERG mode to control pacing consistency, and calibrating challenge levels using initial baseline testing. By focusing on inclusive, data-driven design and comparing distinct social interaction modes, the system seeks to provide more meaningful engagement, better performance outcomes, and deeper insight into how different users respond to varied exergaming experiences.

3. Methodology / Approach / Specification, Design, and implementation
    - 3.1. System specification and Functional Design
        - 3.1.1. Goals for the VR exergame
        - 3.1.2. Description of the three scenarios
        - 3.1.3. Consistent environmental settings.
        - 3.1.4. Performance tracking elements.
    - 3.2. Technical Development and Tools
        - 3.2.1. Development environment (Unity, VR hardware, exercise bike integration).
        - 3.2.2. AI development for adaptive behaviour.
        - 3.2.3. UI/UX considerations and gamification elements (progress bars, post race analytics).
        - 3.2.4. Data logging and analytics pipeline.
        - 3.2.5. Tools/libraries for data collection and visualisation (Python, Unity plugins)
    - 3.3. Methodology for User Study
        - 3.3.1. Participant recruitment, consent, and ethics
        - 3.3.2. Study design (within-subject, counterbalanced if needed).
        - 3.3.3. Metrics:
            - 3.3.3.1. Quantitative (speed, cadence, power, HR, time).
            - 3.3.3.2. Qualitative (user experience, engagement, enjoyment (via surveys)
            - 3.3.3.3. Personality data for exploring correlations.
        - 3.3.4. Risk analysis and mitigation strategies (as in initial plan)
        - 3.3.5. Timeline for development testing and study
4. Results and Evaluation
    - 4.1. Quantitative results
        - 4.1.1. Comparative analysis of performance metrics across the three scenarios
        - 4.1.2. Visualisations (graphs of time, power output, HR, etc)
        - 4.1.3. Statistical Analysis (ANOVA, t-tests for paired comparisons)

4.1.4. Interpretation of results regarding performance under cooperative/competitive dynamics.

4.2. Qualitative Results

4.2.1. Thematic analysis of user feedback (motivation, engagement, enjoyment)

4.2.2. Reflections on user preferences between cooperation and competition.

4.2.3. Summary of personal correlations, if found.

4.3. Evaluation and critical reflection

4.3.1. Effectiveness of gamification and AI in enhancing engagement.

4.3.2. Success and limitations of AI difficulty adjustment

4.3.3. Discussion of accessibility and usability feedback from participants.

4.3.4. Issues encountered (bugs, AI behaviour, data collection reliability).

4.3.5. Validation of the game as a fitness tool.

5. Conclusions and Future Work

5.1. Summary of Key Findings

5.1.1. Impact of competition and cooperation on user performance and engagement.

5.1.2. Strengths and weaknesses of the developed system.

5.2. Contributions to the field

5.2.1. Novelty of integrating social mechanics dynamically in VR exergames.

5.2.2. Insights into VR exergaming for different personality and fitness types.

5.3. Future Work

5.3.1. Improving AI to better adapt to different skill levels.

5.3.2. Expanding to multiplayer online modes.

5.3.3. Longitudinal studies on fitness adherence using the developed platform.

5.3.4. Broader application in rehab and sports training.

References:

Interactive Feedforward in High Intensity VR Exergaming
Chinnachamy Barathi, S. (Author). 16 Sept 2020